RESEARCH LETTER

Does channel attention help few-label hyperspectral CNNs? Gains are confined to classes with one or two training pixels

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ABSTRACT
Channel-attention blocks are often added to lightweight networks for hyperspectral image classification, usually with gains reported on random pixel splits. We test when the choice of block matters. Seven modulation blocks (SE, ECA, CBAM, GCT, SRM and two variants of a dispersion-aware block we introduce, SNSM) are compared on one backbone with one training protocol, no test-based model selection, up to fifteen paired seeds, bootstrap confidence intervals and Holm-corrected Wilcoxon tests. On random 5% splits of Indian Pines, whose three smallest classes then have one or two training pixels, the two gates that use channel dispersion (SNSM, SRM) exceed the other blocks by 1.8–3.8 points of average accuracy (AA), while overall accuracy differs by under 0.2 points. Giving every class 15–50 training pixels, in a random or the standard split, removes this advantage. On patch-disjoint splits of Pavia University and Kennedy Space Center (KSC), accuracy falls by 32–34 points and most differences are not significant. We also show that the distributed KSC file stores small negative values as unsigned integers; reading them as signed raises seven convolutional variants from 77–92% to over 99% AA.

KEYWORDS
Hyperspectral image classification; channel attention; spatial leakage; evaluation protocol; few labels

1. Introduction

Hyperspectral images record hundreds of contiguous bands. This makes pixel-level land-cover classification possible, but when labelled samples are few it exposes classifiers to the Hughes phenomenon (Hughes 1968; Bioucas-Dias et al. 2013). Convolutional neural networks (CNNs) that operate on spatial–spectral patches are the usual lightweight choice (Li et al. 2019; Audebert, Le Saux, and Lefevre 2019; Makantasis et al. 2015; Chen et al. 2016; Zhong et al. 2018; Roy et al. 2020), and transformers add global context at a higher cost (Hong et al. 2022; Sun et al. 2022). Channel attention is a cheap way to make such networks input-dependent: squeeze-and-excitation (SE) predicts one scale per channel from pooled means through fully connected (FC) layers (Hu, Shen, and Sun 2018); ECA replaces the FC layers by a one-dimensional convolution (Wang et al. 2020); CBAM adds a spatial gate (Woo et al. 2018); GCT normalises channel statistics and gates them with $1 + \tanh(\cdot)$ (Yang et al. 2020); and

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SRM pools channel means and standard deviations (Lee, Kim, and Nam 2019). Hyperspectral networks have adopted these blocks widely (Mei et al. 2019; Haut et al. 2019; Li et al. 2020; Zhu et al. 2021; Roy et al. 2021).

Most reported gains come from random pixel splits. With patch-based models, training and test patches then overlap, so test accuracy partly measures memorised spatial context (Liang et al. 2017; Nalepa, Myller, and Kawulok 2019). This leakage has been surveyed (Feng et al. 2023), disjoint sampling changes the accuracy of deep models (Ahmad et al. 2024), and spatially separated splits lower the overall accuracy of CNNs and transformers on Pavia University by about 40 points (Alkhatib 2026). These studies compare whole architectures. They do not show whether the *choice of channel-attention block* matters, or under which conditions.

Our contributions are:

- a controlled comparison of seven modulation blocks on one backbone, with identical inputs and schedule, no test-based model selection, paired seeds, bootstrap intervals and Holm-corrected Wilcoxon tests, under four protocols: random 5% splits, random splits with 15–50 training pixels per class, the standard Indian Pines split, and patch-disjoint splits in which no test patch overlaps a training patch (Sections 2–3);
- the finding that block choice matters only when the smallest classes have one or two training pixels: a random split with the standard split's class counts removes the advantage as completely as the standard split does, so the effect is one of class counts rather than of split geometry (Section 4);
- SNSM, a dispersion-aware block used here as a probe; its ablation shows that the channel variance, which SRM also uses, carries the gain;
- a diagnosis of the Kennedy Space Center (KSC) data encoding, which earlier work treated as sensor saturation (Kakde et al. 2018), and its effect on CNN accuracy.

2. Blocks compared

All blocks are inserted in the same backbone. A 3×3 convolution maps the B input features to $C = 64$ channels (batch normalisation (Ioffe and Szegedy 2015), ReLU). Two residual blocks (He et al. 2016) follow, each computing

$$\mathbf{h} = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{3\times3}(\mathbf{x}))), \qquad \mathbf{y} = \mathrm{ReLU}\big(\mathrm{BN}(\mathrm{Conv}_{3\times3}(\mathcal{M}(\mathbf{h}))) + \mathbf{x}\big), \quad (1)$$

where $\mathcal{M}$ is the block under study ($\mathcal{M}$ = identity for the plain CNN). Global average pooling, dropout (0.4) (Srivastava et al. 2014) and a linear layer produce the class scores. SE, ECA (kernel 3, the adaptive value for 64 channels) and CBAM (reduction 16, 7×7 spatial gate) output a sigmoid scale per channel; GCT uses its ℓ_2 embedding and an identity start; SRM gates each channel from its mean and standard deviation through a channel-wise FC layer and batch normalisation.

SNSM. For $\mathbf{h} \in \mathbb{R}^{C\times H\times W}$, SNSM computes for every channel c the spatial mean μ_c, the spatial variance v_c and the cross-channel gradient $g_c = (\mu_{c+1} - \mu_{c-1})/2$. Each descriptor vector is standardised across channels, the three are stacked into $\mathbf{S} \in \mathbb{R}^{3\times C}$, and two one-dimensional convolutions along the channel axis (8 hidden channels, kernel

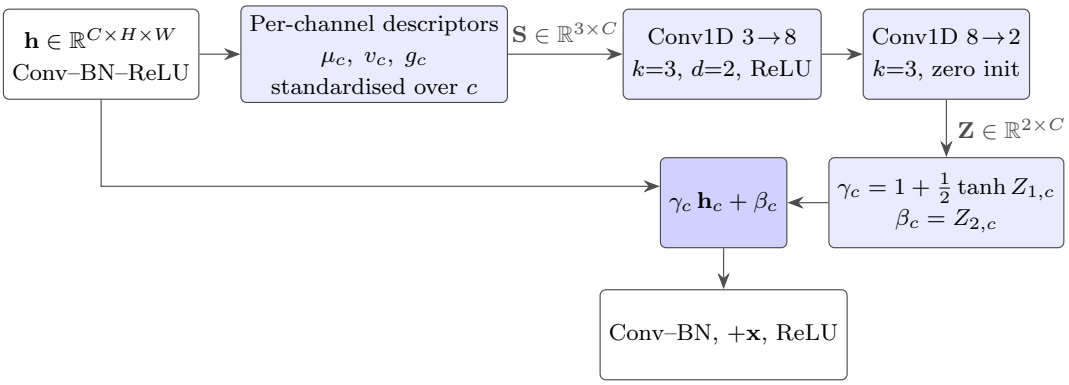


Figure 1. SNSM inside a residual block. Per-channel descriptors of the feature map $\mathbf{h}$ are filtered along the channel axis by two one-dimensional convolutions, whose output sets a bounded scale γ_c and a shift β_c for each channel. The second convolution starts at zero, so the block is initially the identity.

3, dilation 2 in the first, replicate padding) give $\mathbf{Z} \in \mathbb{R}^{2\times C}$. The modulation is

$$\gamma_c = 1 + \tfrac{1}{2}\tanh(Z_{1,c}), \qquad \beta_c = Z_{2,c}, \qquad \mathcal{M}(\mathbf{h})_c = \gamma_c\,\mathbf{h}_c + \beta_c. \tag{2}$$

The second convolution starts at zero, so the block is initially the identity; it adds 130 parameters per block (Figure 1). SNSM (γ) drops the shift. Like SRM it sees channel dispersion; unlike SRM it filters the descriptors jointly across neighbouring feature channels and can shift as well as scale. Including it with and without each descriptor lets us test which channel statistic matters.

3. Data and protocols

Data. We use Indian Pines (IP), Pavia University (PU), KSC and Botswana (BW) (Baumgardner, Biehl, and Landgrebe 2015; Gamba 2004; Grupo de Inteligencia Computacional 2020) (Table 1). Each band is smoothed with a 5×5 Wiener filter, principal component analysis (PCA) (Jolliffe and Cadima 2016), fitted without labels on all pixels, keeps 30 components, and the db4 wavelet approximation of each component (Mallat 1989) is resized and appended, giving $B = 60$ features. Each labelled pixel is represented by its $15 \times 15 \times B$ patch.

KSC encoding. The commonly distributed KSC file is stored as unsigned 16-bit integers, but it contains small negative values (-27 to -1) that are stored in two's complement and are therefore read as 65,509–65,535; about 5% of its pixels contain at least one such value. Kakde et al. (2018) noticed these values, described them as saturation and set them to zero. The DeepHyperX toolbox accompanying Audebert, Le Saux, and Lefevre (2019) loads the file unchanged and scales it by its global minimum and maximum, which compresses the valid range of 0–1,244 into less than 2% of the scale. We did not audit which published KSC results used such a loader. We read KSC as signed integers. Without this correction, the wrapped values dominate the PCA and seven CNN variants reach only 77–92% AA on KSC with random 5% splits, instead of 99.2–99.5%.

Protocols. *Random 5%*: for each class, 5% of the labelled pixels (at least one) train and the rest test, giving 505/9,744 (IP), 2,135/40,641 (PU), 256/4,955 (KSC) and 156/3,092 (BW) pixels; Alfalfa, Grass-pasture-mowed and Oats then have one or two training pixels. *Random 50/15* (IP): a random split with the class counts of

Table 1. Benchmark scenes and the random 5% splits (seed 42; split sizes are the same for every seed).

Scene	Sensor	Size (pixels)	Bands	GSD (m)	Classes	Labelled	Train/test (random)
Indian Pines (IP)	AVIRIS	145×145	200	20	16	10,249	505/9,744
Pavia University (PU)	ROSIS	610×340	103	1.3	9	42,776	2,135/40,641
Kennedy Space Center (KSC)	AVIRIS	512×614	176	18	13	5,211	256/4,955
Botswana (BW)	Hyperion	1476×256	145	30	14	3,248	156/3,092



Figure 2. Patch-disjoint split of Pavia University (seed 42). Training pixels form one compact cluster per class; test pixels whose 15×15 patch would overlap a training patch are excluded.

the standard split, 15 training pixels for those three classes and 50 for each of the others (695/9,554). *Standard* (IP): the fixed training and test sets distributed with SpectralFormer (Hong et al. 2022) (695/9,671). Its training pixels come from separate regions, but with 15×15 patches every test patch still overlaps a training patch; the corresponding PU and Houston sets were no longer available from the authors' link. *Patch-disjoint* (PU, KSC): for each class, the 5% of its pixels nearest a random centre form a compact training cluster, and every test pixel within 14 pixels of any training pixel is discarded, so no test patch overlaps a training patch (Figure 2); 33,847–36,453 (PU) and 3,077–3,513 (KSC) test pixels remain, covering all 9 PU classes and at least 11 of the 13 KSC classes. The same rule leaves no test pixels for five IP classes, so IP is not evaluated this way.

Training and statistics. Every model sees the same features and splits for a given seed. Training uses cross-entropy, AdamW (Loshchilov and Hutter 2019) (learning rate 10^{-3}, weight decay 10^{-4}), step decay (factor 0.5 every 40 epochs), batch size 64 and exactly 100 epochs, without augmentation or early stopping; the final model is evaluated once, so no validation or test data are used for selection. All eight backbone variants use 15 seeds (42–56) under the two IP random protocols, the standard split and the patch-disjoint splits, and five seeds on the other random splits. On the standard

Table 2. Random 5% splits: OA and AA (%, mean ± std). The eight backbone variants use 15 seeds on IP and five elsewhere; FC (γ+β), HybridSN (Roy et al. 2020) and SSFTT (Sun et al. 2022) use five. Best mean per column in bold.

Method	IP		KSC		PU		BW	
	OA	AA	OA	AA	OA	AA	OA	AA
Plain CNN	96.28±0.79	89.49±2.44	99.65±0.34	99.51±0.36	**99.59±0.07**	99.01±0.16	99.09±0.55	98.87±0.90
SE (γ)	96.34±0.62	90.04±2.18	99.61±0.28	99.43±0.38	99.53±0.19	98.80±0.56	98.95±0.82	98.76±1.07
ECA (γ)	96.28±0.68	90.87±2.53	99.61±0.47	99.40±0.66	99.58±0.10	99.00±0.20	**99.13±0.76**	**98.90±1.08**
CBAM	96.19±0.60	91.58±1.49	99.54±0.56	99.23±0.83	99.58±0.06	**99.11±0.18**	98.99±0.87	98.78±1.20
GCT	96.28±0.80	90.07±2.34	**99.69±0.26**	**99.55±0.28**	99.58±0.08	98.99±0.18	99.11±0.56	98.88±0.91
SRM	96.20±0.71	93.04±2.14	99.69±0.36	99.51±0.49	99.55±0.07	98.92±0.17	98.87±0.63	98.67±0.97
SNSM (γ)	96.17±0.73	92.89±2.00	99.69±0.26	99.50±0.39	99.49±0.05	98.91±0.16	99.10±0.62	98.89±0.96
SNSM (γ+β)	96.27±0.67	**93.33±1.68**	99.63±0.55	99.45±0.73	99.47±0.05	98.93±0.16	99.04±0.69	98.85±1.03
FC (γ+β)	96.24±0.52	92.51±1.60	99.63±0.40	99.48±0.47	99.51±0.06	98.88±0.14	98.80±0.70	98.63±0.98
HybridSN	88.30±4.91	73.64±8.28	95.00±3.02	91.20±5.32	97.84±1.52	95.46±4.12	77.16±37.92	75.56±38.31
SSFTT	**96.71±0.79**	92.35±1.81	99.38±0.28	99.15±0.32	–	–	–	–

Table 3. Indian Pines: AA (%, mean ± std over 15 seeds) of the eight backbone variants under three protocols. The last row is the range of the block means.

Block	Random 5%	Random, 50/15 per class	Standard split
Plain CNN	89.49±2.44	97.83±0.52	97.24±0.23
SE (γ)	90.04±2.18	**97.90±0.44**	97.25±0.26
ECA (γ)	90.87±2.53	97.79±0.49	97.15±0.22
CBAM	91.58±1.49	97.78±0.51	97.45±0.29
GCT	90.07±2.34	97.82±0.53	97.23±0.22
SRM	93.04±2.14	97.76±0.57	**97.50±0.22**
SNSM (γ)	92.89±2.00	97.77±0.54	97.37±0.22
SNSM (γ+β)	**93.33±1.68**	97.82±0.43	97.48±0.26
Range of block means	3.84	0.14	0.35

split the seed changes only the initialisation and batch order; elsewhere it also changes the split. For each scene and protocol we compare SNSM (γ+β) with the seven other blocks by paired same-seed differences, report a 95% percentile bootstrap interval of the mean difference, and apply two-sided Wilcoxon signed-rank tests with Holm correction over those seven comparisons. We call blocks practically indistinguishable when their mean AA differs by less than 0.5 points. Runs used PyTorch (Paszke et al. 2019) on NVIDIA Tesla T4 GPUs, except ten KSC patch-disjoint seeds, which ran on an RTX 3050 laptop GPU; each paired comparison uses runs from the same GPU.

4. Results

Random 5% splits. On PU, KSC and BW the eight backbone variants lie within 0.32 AA points of each other (Table 2). Only IP separates them: SNSM (γ+β) has the highest AA and is significantly better than the plain CNN, SE, ECA, CBAM and GCT, by 1.75–3.84 points (Table 4), but not than SRM or SNSM (γ). The overall accuracy of all eight variants differs by only 0.17 points, so the gain comes from a few classes. Oats and Grass-pasture-mowed have one training pixel and 19 and 27 test pixels, so a few test pixels move their per-class accuracy by tens of points. HybridSN and SSFTT, trained under our protocol, are less accurate than the backbone variants.

Class counts, not split geometry. When every IP class has 15–50 training pixels, the advantage disappears (Table 3). In the random 50/15 split the eight block means span only 0.14 AA points, no difference is significant and every bootstrap interval lies within ±0.3 points. In the standard split they span 0.35 points; the only significant difference, against ECA, is 0.33 points. Because the random 50/15 split keeps the random geometry and changes only the class counts, the 3.84-point spread of the

Table 4. Paired AA differences SNSM (γ+β) − block (points) with 95% bootstrap intervals. *: Holm-corrected two-sided Wilcoxon $p < 0.05$ (Holm family: the seven comparisons in one column).

SNSM (γ+β) −	IP random 5%	IP random 50/15	IP standard	PU disjoint	KSC disjoint
Plain CNN	+3.84* [+2.6, +5.1]	-0.01 [-0.2, +0.2]	+0.24 [+0.1, +0.4]	+0.31 [-1.6, +2.2]	+3.46* [+1.7, +5.1]
SE (γ)	+3.29* [+2.4, +4.3]	-0.08 [-0.2, +0.0]	+0.23 [+0.1, +0.4]	-0.47 [-2.5, +1.4]	-0.19 [-2.1, +1.6]
ECA (γ)	+2.45* [+1.1, +3.9]	+0.04 [-0.1, +0.2]	+0.33* [+0.2, +0.5]	-0.39 [-2.2, +1.4]	+1.86 [+0.6, +3.3]
CBAM	+1.75* [+1.0, +2.5]	+0.05 [-0.1, +0.2]	+0.04 [-0.1, +0.2]	+1.03 [-0.6, +2.4]	+0.91 [-0.7, +2.4]
GCT	+3.26* [+2.2, +4.4]	+0.01 [-0.2, +0.2]	+0.25 [+0.1, +0.4]	+0.92 [-0.8, +2.5]	+3.51* [+1.8, +5.2]
SRM	+0.29 [-0.6, +1.3]	+0.07 [-0.1, +0.2]	-0.02 [-0.2, +0.1]	+1.33 [-0.3, +2.7]	+4.25* [+2.0, +6.6]
SNSM (γ)	+0.44 [-0.2, +1.1]	+0.06 [-0.1, +0.3]	+0.12 [-0.1, +0.3]	+0.03 [-1.7, +1.8]	+1.40 [-0.7, +3.6]

Table 5. PU and KSC: AA under random 5% splits (five seeds) and OA and AA under patch-disjoint splits (15 seeds) (%, mean ± std). The last row is the range of the block means.

Block	PU			KSC		
	AA, random	OA, disjoint	AA, disjoint	AA, random	OA, disjoint	AA, disjoint
Plain CNN	99.01±0.16	55.31±9.82	67.03±5.03	99.51±0.36	75.13±4.97	63.55±6.41
SE (γ)	98.80±0.56	55.91±9.73	**67.81±4.39**	99.43±0.38	76.79±4.44	**67.20±6.66**
ECA (γ)	99.00±0.20	56.30±9.17	67.73±4.43	99.40±0.66	76.40±4.81	65.15±5.86
CBAM	**99.11±0.18**	56.11±10.31	66.31±4.80	99.23±0.83	76.68±4.74	66.10±7.39
GCT	98.99±0.18	54.59±9.68	66.42±4.57	**99.55±0.28**	75.20±5.16	63.49±6.80
SRM	98.92±0.17	56.02±9.99	66.00±4.73	99.51±0.49	74.36±4.22	62.75±6.38
SNSM (γ)	98.91±0.16	55.41±9.44	67.31±4.57	99.50±0.39	76.86±4.19	65.60±7.15
SNSM (γ+β)	98.93±0.16	**56.80±11.83**	67.34±4.95	99.45±0.73	**77.11±3.54**	67.01±6.00
Range of block means	0.31	2.21	1.80	0.32	2.75	4.44

random 5% protocol is explained by the one or two training pixels of the smallest classes, not by spatial leakage.

Patch-disjoint splits. Removing patch overlap lowers AA by 32.0 points on PU and 34.3 on KSC, averaged over blocks, and OA to 55–57% and 74–77% (Table 5). Block means now span 1.80 (PU) and 4.44 (KSC) points, but the seed-to-seed standard deviation is 4.4–7.4 points and the median bootstrap interval is 3.5 (PU) and 3.4 (KSC) points wide, so differences of about two points can be neither confirmed nor excluded. On PU no difference is significant; on KSC, SNSM (γ+β) is significantly better than the plain CNN, GCT and SRM, but not than SE, ECA, CBAM or SNSM (γ), and SE has the highest mean AA (Figure 3). The ranking of blocks also changes between protocols: SRM, tied best on random 5% IP, is the lowest on patch-disjoint KSC.

Which statistic matters. Removing descriptors from SNSM on random 5% IP (five seeds; Table 6) shows that the variance carries the gain: with the mean alone AA falls by 1.16 points, and adding the gradient to the mean does not recover it; without the zero initialisation AA is lower on all five seeds. This matches the tie between SNSM and SRM, the only other block that gates on dispersion, and suggests that the spread of a channel's response helps when a class is represented by one or two pixels.

Cost. The channel blocks add between 6 (ECA) and 4,256 (SE) parameters to the plain CNN, and SNSM adds 260 (Table 7). SNSM's descriptor computation consists of many small operations, so its batch-1 latency is about 3.7 times and its training time about 1.7 times that of the plain CNN.

5. Discussion and limitations

Across four scenes and four protocols, the choice of channel-attention block produced large and consistent differences in only one setting: random 5% splits of Indian Pines, where three classes have one or two training pixels. There, the two gates that see channel dispersion were clearly better than SE, ECA, CBAM and GCT; elsewhere the block means differed by less than 0.5 AA points or, on the patch-disjoint splits, by

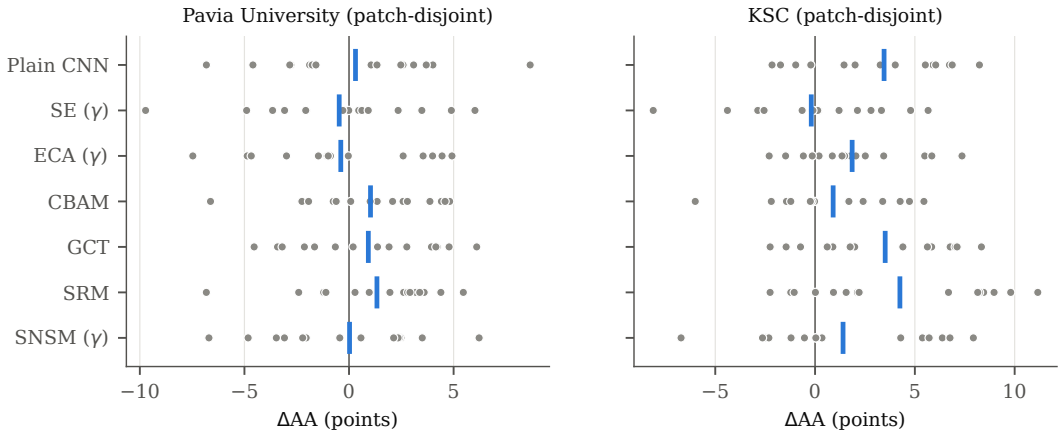


Figure 3. Paired same-seed AA differences SNSM (γ+β) − block on the patch-disjoint splits. Grey dots are the 15 seeds; the blue bar is their mean.

Table 6. SNSM (γ+β) ablation on IP, random 5% splits (AA, %; five seeds). ΔAA is the paired difference from the full block; wins count seeds on which the variant is better.

Descriptors	AA	ΔAA (wins/5)
mean, var, grad (SNSM)	94.16±1.07	–
mean	93.01±0.72	-1.16 (1/5)
mean, var	94.55±1.67	+0.39 (4/5)
mean, grad	93.37±1.09	-0.80 (1/5)
mean, var, grad; default init.	92.67±1.15	-1.49 (0/5)

amounts mostly within seed-to-seed noise. The matched-count control shows that this is an effect of extreme class scarcity, not of spatial leakage. Leakage instead changes the level of accuracy: once no test patch overlapped a training patch, every block lost more than 30 AA points, and most block differences were no longer significant. Gains of a few AA points reported for a new block should therefore be checked against the number of training pixels in the smallest classes.

Several limitations apply. The patch-disjoint construction trains on one compact cluster per class, so it adds a spatial shift between training and test data as well as removing overlap, and it leaves few test pixels for some KSC classes; the standard IP split still lets patches overlap, and the standard PU and Houston splits were unavailable. PCA is fitted on all pixels of each scene without labels. All blocks share one shallow backbone, one patch size, one feature pipeline and a fixed schedule; none was tuned, and deeper networks or transformers may use attention differently. HybridSN and SSFTT are less accurate here than in their original papers. With 15 seeds, the patch-disjoint comparisons can only exclude differences larger than about two points.

6. Conclusion

For lightweight few-label hyperspectral CNNs, the choice among seven channel-attention blocks mattered only when the smallest classes had one or two training pixels. In that setting, dispersion-based gates, including the proposed SNSM, added 2–4 AA points; with 15–50 training pixels per class, in a random or the standard split, all blocks were practically indistinguishable, and under patch-disjoint splits, where

Table 7. Model size and cost for a $15 \times 15 \times 60$ IP patch. Training time is the mean per run (100 epochs, random 5% splits).

Method	Params (M)	MMACs[a]	Latency (ms)[b]	Train IP (s)	Train PU (s)
Plain CNN	0.184	41.2	0.69	4.2	17.6
SE (γ)	0.188	41.2	1.17	4.5	19.3
ECA (γ)	0.184	41.2	0.98	4.8	19.8
CBAM	0.185	41.3	1.55	5.9	24.9
GCT	0.184	41.2	1.11	4.7	20.0
SRM	0.184	41.2	1.22	4.8	20.2
SNSM (γ)	0.184	41.3	2.54	7.1	29.5
SNSM (γ+β)	0.184	41.3	2.54	7.2	30.2
FC (γ+β)	0.190	41.2	1.31	4.7	19.8
HybridSN	1.743	137.4	0.71	26.2	115.3
SSFTT	0.291	34.9	1.25	7.6	–

[a] Multiply–accumulates of convolution and linear layers counted by `thop`; tensor operations such as attention products and descriptor statistics are not included. [b] Batch size 1 on a Tesla T4.

accuracy dropped by 32–34 points, most differences were not significant. Class counts, split design and data handling, such as the KSC encoding corrected here, affect reported accuracy far more than the attention block.

Disclosure statement

The authors report there are no competing interests to declare.

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Data and code availability

The datasets are publicly available (Grupo de Inteligencia Computacional 2020); the standard IP split is distributed with SpectralFormer (Hong et al. 2022). Code, per-run results and the scripts that generate every table and figure are available at `https://github.com/Devguru-codes/research_paper_code`.

Use of generative AI

Generative AI (Claude, Anthropic) was used to assist with writing experiment code, checking references and editing the manuscript text. The authors reviewed and verified all code, results and text, and take full responsibility for the content.

References

Ahmad, Muhammad, Manuel Mazzara, Salvatore Distefano, Adil Mehmood Khan, and Hamad Ahmed Altuwaijri. 2024. "Improving Generalization for Hyperspectral Image Classification: The Impact of Disjoint Sampling on Deep Models." *Computers, Materials & Continua* 81 (1): 503–532. https://doi.org/10.32604/cmc.2024.056318.

Alkhatib, Mohammed Q. 2026. "Data Leakage in Patch-Based Hyperspectral Image Classification: Quantifying the Impact of Spatial Overlap." arXiv preprint arXiv:2610.08770. https://doi.org/10.48550/arXiv.2610.08770.

Audebert, Nicolas, Bertrand Le Saux, and Sebastien Lefevre. 2019. "Deep Learning for Classification of Hyperspectral Data: A Comparative Review." *IEEE Geoscience and Remote Sensing Magazine* 7 (2): 159–173. https://doi.org/10.1109/MGRS.2019.2912563.

Baumgardner, Marion F., Larry L. Biehl, and David A. Landgrebe. 2015. "220 Band AVIRIS Hyperspectral Image Data Set: June 12, 1992 Indian Pine Test Site 3." Purdue University Research Repository. https://doi.org/10.4231/R7RX991C.

Bioucas-Dias, Jose M., Antonio Plaza, Gustavo Camps-Valls, Paul Scheunders, Nasser Nasrabadi, and Jocelyn Chanussot. 2013. "Hyperspectral Remote Sensing Data Analysis and Future Challenges." *IEEE Geoscience and Remote Sensing Magazine* 1 (2): 6–36. https://doi.org/10.1109/MGRS.2013.2244672.

Chen, Yushi, Hanlu Jiang, Chunyang Li, Xiuping Jia, and Pedram Ghamisi. 2016. "Deep Feature Extraction and Classification of Hyperspectral Images Based on Convolutional Neural Networks." *IEEE Transactions on Geoscience and Remote Sensing* 54 (10): 6232–6251. https://doi.org/10.1109/TGRS.2016.2584107.

Feng, Hao, Yongcheng Wang, Zheng Li, Ning Zhang, Yuxi Zhang, and Yunxiao Gao. 2023. "Information Leakage in Deep Learning-Based Hyperspectral Image Classification: A Survey." *Remote Sensing* 15 (15): 3793. https://doi.org/10.3390/rs15153793.

Gamba, Paolo. 2004. "ROSIS Pavia University Hyperspectral Data Set." Telecommunications and Remote Sensing Laboratory, University of Pavia. `https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes`.

Grupo de Inteligencia Computacional. 2020. "Hyperspectral Remote Sensing Scenes." University of the Basque Country (UPV/EHU). `https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes`.

Haut, Juan Mario, Mercedes E. Paoletti, Javier Plaza, Antonio Plaza, and Jun Li. 2019. "Visual Attention-Driven Hyperspectral Image Classification." *IEEE Transactions on Geoscience and Remote Sensing* 57 (10): 8065–8080. https://doi.org/10.1109/TGRS.2019.2918080.

He, Kaiming, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. "Deep Residual Learning for Image Recognition." In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778. https://doi.org/10.1109/CVPR.2016.90.

Hong, Danfeng, Zhu Han, Jing Yao, Lianru Gao, Bing Zhang, Antonio Plaza, and Jocelyn Chanussot. 2022. "SpectralFormer: Rethinking Hyperspectral Image Classification With Transformers." *IEEE Transactions on Geoscience and Remote Sensing* 60: 1–15. https://doi.org/10.1109/TGRS.2021.3130716.

Hu, Jie, Li Shen, and Gang Sun. 2018. "Squeeze-and-Excitation Networks." In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 7132–7141. https://doi.org/10.1109/CVPR.2018.00745.

Hughes, G. 1968. "On the mean accuracy of statistical pattern recognizers." *IEEE Transactions on Information Theory* 14 (1): 55–63. https://doi.org/10.1109/TIT.1968.1054102.

Ioffe, Sergey, and Christian Szegedy. 2015. "Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift." In *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, 448–456. https://doi.org/10.48550/arXiv.1502.03167.

Jolliffe, Ian T., and Jorge Cadima. 2016. "Principal component analysis: a review and recent developments." *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 374 (2065): 20150202. https://doi.org/10.1098/rsta.2015.0202.

Kakde, Deovrat, Arin Chaudhuri, Hansi Jiang, Carol Sadek, Seunghyun Kong, and Yuwei Liao. 2018. "A New Bandwidth Selection Criterion for Using SVDD to Analyze Hyperspectral Data." In *Algorithms and Technologies for Multispectral, Hyperspectral, and Ultraspectral Imagery XXIV*, Proc. SPIE. https://doi.org/10.1117/12.2314964.

Lee, Hyunjae, Hyo-Eun Kim, and Hyeonseob Nam. 2019. "SRM: A Style-Based Recalibration Module for Convolutional Neural Networks." In *2019 IEEE/CVF International Conference*

on Computer Vision (ICCV), 1854–1862. https://doi.org/10.1109/ICCV.2019.00194.

Li, Rui, Shunyi Zheng, Chenxi Duan, Yang Yang, and Xiqi Wang. 2020. "Classification of Hyperspectral Image Based on Double-Branch Dual-Attention Mechanism Network." *Remote Sensing* 12 (3): 582. https://doi.org/10.3390/rs12030582.

Li, Shutao, Weiwei Song, Leyuan Fang, Yushi Chen, Pedram Ghamisi, and Jon Atli Benediktsson. 2019. "Deep Learning for Hyperspectral Image Classification: An Overview." *IEEE Transactions on Geoscience and Remote Sensing* 57 (9): 6690–6709. https://doi.org/10.1109/TGRS.2019.2907932.

Liang, Jie, Jun Zhou, Yuntao Qian, Lian Wen, Xiao Bai, and Yongsheng Gao. 2017. "On the Sampling Strategy for Evaluation of Spectral-Spatial Methods in Hyperspectral Image Classification." *IEEE Transactions on Geoscience and Remote Sensing* 55 (2): 862–880. https://doi.org/10.1109/TGRS.2016.2616489.

Loshchilov, Ilya, and Frank Hutter. 2019. "Decoupled Weight Decay Regularization." In *International Conference on Learning Representations (ICLR)*, https://doi.org/10.48550/arXiv.1711.05101.

Makantasis, Konstantinos, Konstantinos Karantzalos, Anastasios Doulamis, and Nikolaos Doulamis. 2015. "Deep supervised learning for hyperspectral data classification through convolutional neural networks." In *2015 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 4959–4962. https://doi.org/10.1109/IGARSS.2015.7326945.

Mallat, S.G. 1989. "A theory for multiresolution signal decomposition: the wavelet representation." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 11 (7): 674–693. https://doi.org/10.1109/34.192463.

Mei, Xiaoguang, Erting Pan, Yong Ma, Xiaobing Dai, Jun Huang, Fan Fan, Qinglei Du, Hong Zheng, and Jiayi Ma. 2019. "Spectral-Spatial Attention Networks for Hyperspectral Image Classification." *Remote Sensing* 11 (8): 963. https://doi.org/10.3390/rs11080963.

Nalepa, Jakub, Michal Myller, and Michal Kawulok. 2019. "Validating Hyperspectral Image Segmentation." *IEEE Geoscience and Remote Sensing Letters* 16 (8): 1264–1268. https://doi.org/10.1109/LGRS.2019.2895697.

Paszke, Adam, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, et al. 2019. "PyTorch: An Imperative Style, High-Performance Deep Learning Library." In *Advances in Neural Information Processing Systems 32 (NeurIPS)*, 8024–8035. https://doi.org/10.48550/arXiv.1912.01703.

Roy, Swalpa Kumar, Gopal Krishna, Shiv Ram Dubey, and Bidyut B. Chaudhuri. 2020. "HybridSN: Exploring 3-D–2-D CNN Feature Hierarchy for Hyperspectral Image Classification." *IEEE Geoscience and Remote Sensing Letters* 17 (2): 277–281. https://doi.org/10.1109/LGRS.2019.2918719.

Roy, Swalpa Kumar, Suvojit Manna, Tiecheng Song, and Lorenzo Bruzzone. 2021. "Attention-Based Adaptive Spectral–Spatial Kernel ResNet for Hyperspectral Image Classification." *IEEE Transactions on Geoscience and Remote Sensing* 59 (9): 7831–7843. https://doi.org/10.1109/TGRS.2020.3043267.

Srivastava, Nitish, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. 2014. "Dropout: A Simple Way to Prevent Neural Networks from Overfitting." *Journal of Machine Learning Research* 15 (56): 1929–1958. `https://jmlr.org/papers/v15/srivastava14a.html`.

Sun, Le, Guangrui Zhao, Yuhui Zheng, and Zebin Wu. 2022. "Spectral–Spatial Feature Tokenization Transformer for Hyperspectral Image Classification." *IEEE Transactions on Geoscience and Remote Sensing* 60: 1–14. https://doi.org/10.1109/TGRS.2022.3144158.

Wang, Qilong, Banggu Wu, Pengfei Zhu, Peihua Li, Wangmeng Zuo, and Qinghua Hu. 2020. "ECA-Net: Efficient Channel Attention for Deep Convolutional Neural Networks." In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 11531–11539. https://doi.org/10.1109/CVPR42600.2020.01155.

Woo, Sanghyun, Jongchan Park, Joon-Young Lee, and In So Kweon. 2018. "CBAM: Convolutional Block Attention Module." In *Computer Vision – ECCV 2018, Lecture Notes in Computer Science, vol. 11211*, 3–19. https://doi.org/10.1007/978-3-030-01234-2_1.

Yang, Zongxin, Linchao Zhu, Yu Wu, and Yi Yang. 2020. "Gated Channel Transformation for Visual Recognition." In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 11791–11800. https://doi.org/10.1109/CVPR42600.2020.01181.

Zhong, Zilong, Jonathan Li, Zhiming Luo, and Michael Chapman. 2018. "Spectral–Spatial Residual Network for Hyperspectral Image Classification: A 3-D Deep Learning Framework." *IEEE Transactions on Geoscience and Remote Sensing* 56 (2): 847–858. https://doi.org/10.1109/TGRS.2017.2755542.

Zhu, Minghao, Licheng Jiao, Fang Liu, Shuyuan Yang, and Jianing Wang. 2021. "Residual Spectral–Spatial Attention Network for Hyperspectral Image Classification." *IEEE Transactions on Geoscience and Remote Sensing* 59 (1): 449–462. https://doi.org/10.1109/TGRS.2020.2994057.